CellHack2

Quick start Rules Reference About

Rules

Setup

The arena is a grid, 64×64 on the ladder. Four colonies start with one cell each, 10 energy. Outside the grid is wall.

Each tick

1. **Decide.** Every cell's function is called with a snapshot of the world and returns one action. Nobody sees anyone else's choice. The engine counts the instructions each decision took.
2. **Act, fastest thinker first.** Cells take their turns in order of how many instructions their decision ran, fewest first. Each action sees the board as the turns before it left it.
3. `REST` gains 1.
4. `EAT` takes up to 2 from the cell in that direction and adds it to yours. **The bitten cell is stunned**: if it has not had its turn yet, it loses its action this tick.
5. `FEED` gives 2 of yours (or all you have) to the cell in that direction.
6. `SPLIT` puts a new cell there with half your energy, if the square is empty and you have at least 5.
7. `MOVE` steps there, if the square is empty.
8. A cell that reaches 0 dies on the spot. It takes no turn, nobody can eat or feed it, and it keeps its square until the end of the tick.
9. **Tidy.** Energy is capped at 50. The dead are removed.

Equal instruction counts are ordered at random, so identical code is a coin flip.

A cell whose code faults (crashes, runs more than 1000 instructions, returns an unknown action) does nothing that tick. Faults appear in the match page's log.

Winning

A match ends when the tick limit is reached, when one colony is left, or when all are dead. Living colonies rank by population, then total energy, then fewer instructions run over the whole match. Dead colonies rank below them by how long they lasted. Ladder matches award 3, 2, 1 and 0 points by placement, which feed your rating.

Worth knowing

- Resting is the only income. A tick spent doing anything else is a tick without the 1.
- Instructions cost no energy. They decide the order of turns.
- One eater beats one rester: 2 out against 1 in. A line of resters is eaten through. Eating back, feeding the front, or fat reserves are what hold a line.
- Eating a cell that has 2 or less kills it.
- You can eat and feed your own cells. Eating or feeding an empty square does nothing.
- A split halves your energy. Early splits grow numbers; late splits grow strength.
- The 1000-instruction limit is far above what a simple cell needs: the bots use fewer than 20.

▼ Fine print

- **Order.** One pass per tick, fewest-instructions-first; equal counts are shuffled at random. A cell eaten by several others loses energy to each in turn until it has none.
- **Targets** are whatever is in that square when your turn comes, not when you decided. A bite at a cell that has already stepped away finds nothing; a bite at a square someone stepped into hits the newcomer.
- **Moves.** Two cells cannot swap places. A square vacated earlier in the tick can be entered later in the same tick; a square freed by a death cannot until the next tick.
- **Nothing happens** when you split with less than 5, or split or move into a wall or an occupied square. A feeder with less than 2 gives all it has, and dies if that was all of it.
- **Newborns** act from the next tick, with `VIEW_AGE` 0 and a copy of

the parent's memory.

- **Ties** between living colonies with equal population, energy and instruction count, or dead colonies that died on the same tick, go to the lower colony number.
- **Randomness** comes only from the match seed. `VIEW_RAND` is drawn from it, `rdtsc` returns the tick number, and a match replays identically every time.